⚡ OSTH SPLUNK QUERY ARSENAL V4

All sources merged • 175+ queries • ADCS · NTLM Relay · Silver Ticket · BYOVD · DCOM · RBCD · Kill-Chain Correlation

🔍 Filter (mimikatz, lateral, 4624, akira, lockbit, golden ticket...)

All | Windows | Linux | Web | DC | Database | Mail

JUMP TO SECTION

- 0. First-Minute Triage
- 1. Reconnaissance
- 2. Initial Access
- 3. Execution
- 4. Persistence
- 5. Privilege Escalation
- 6. Defense Evasion
- 7. Credential Access
- 8. Discovery
- 9. Lateral Movement
- 10. C2
- 11. Exfiltration
- 12. Ransomware
- 13. IoC Hunting
- 14. Timeline
- 15. Linux/Web
- 16. Web Attacks
- 17. Kerberos Attacks
- 18. ADCS / Certificate Abuse
- 19. NTLM Relay / Coercion
- 20. Kill-Chain Correlation
- Quick Reference

0. FIRST-MINUTE TRIAGE

11 ▼

Process Overview — All Hosts

ALL

All process creation. Spot unusual binaries.

```
index=* EventCode=1 | stats count by host Image | sort - count
```

Network Connections Overview

ALL

All outbound. Unknown IPs, odd ports.

```
index=* EventCode=3 | stats count by host Image DestinationIp DestinationPort | sort - count
```

PowerShell Script Blocks

WIN

Every PS script block. Encoded cmds, downloads, recon.

```
index=* EventCode=4104 | table _time host Message | sort _time
```

New Account Creation

WIN DC

Rogue: helpdesk_1, support_388945a0, adm1n.

```
index=* EventCode=4720 | table _time host SubjectUserName TargetUserName
```

Scheduled Task Creation

WIN

Persistence via tasks.

```
index=* EventCode=4698 | table _time host SubjectUserName TaskName TaskContent
```

New Service Installation

WIN

bits, PSEXESVC, logwatcher, prst_servshell443.

```
index=* EventCode=7045 | table _time host ServiceName ServiceFileName AccountName
```

NTLM Auth — Attacker Hostname

WIN DC

CRITICAL: Workstation field = attacker machine name.

```
index=* EventCode=4776 | table _time host TargetUserName Workstation | sort _time
```

ALL Known Attacker IPs (Mega)

ALL

Every known attacker/C2 IP from all sources.

```
index=* ("192.168.50.235" OR "192.168.50.11" OR "192.168.12.231" OR "192.168.12.60" OR "192.168.50.63" OR "192.168.50.
67" OR "192.168.51.50" OR "192.168.51.54" OR "192.229.211.108" OR "222.184.79.11" OR "226.211.90.236" OR "172.16.51.33
") | table _time host Image CommandLine DestinationIp DestinationPort | sort _time
```

Ransomware Quick Check

ALL

Notes, encrypted extensions, known families.

```
index=* ("akira_readme" OR ".nmh" OR "lockbit" OR "RECOVER" OR "decrypt" OR "YOUR FILES" OR "petya" OR "mischa" OR "go
ldeneye" OR "ransom") | table _time host _raw | sort _time
```

Failed Logon Burst (Brute Force)

WIN DC

Password spraying / brute force. Training: 0xc000006a, 0xc000006d status codes.

```
index=* EventCode=4625 | stats count by TargetUserName IpAddress WorkstationName | sort - count
```

Top Destination IPs per Host

ALL

Training: Quick C2 beacon discovery — find top talkers per host.

```
index=* EventCode=3 | stats count by SourceHostname DestinationIp | sort - count | head 20
```

🔭 1. RECONNAISSANCE

5 ▼

Port Scanning Tools

ALL

Nmap, masscan, rustscan.

```
index=* ("nmap" OR "masscan" OR "rustscan" OR "zenmap") | table _time host Image CommandLine User
```

Web Dir Brute / Scanner UA

WEB

Scanner tools + suspicious user-agents. Training: Windows XP UA from 192.168.51.54.

```
index=* ("gobuster" OR "dirbuster" OR "nikto" OR "wfuzz" OR "ffuf" OR "dirb" OR "sqlmap" OR "Windows XP" OR "python-re
quests") | table _time host _raw | sort _time
```

HTTP 404 Burst

WEB

High 404 volume = dir scan.

```
index=* "404" ("GET" OR "POST") | stats count by clientip | sort - count
```

SMB Enum (Multi-target 445)

WIN

One source scanning many hosts on 445.

```
index=* EventCode=3 (DestinationPort=445 OR DestinationPort=139) | stats count dc(DestinationIp) as targets by SourceIp Image | sort - targets
```

Suricata / IDS Alerts

WEB

Training: IDS1/snort01 run Suricata. Check for ET MALWARE/TROJAN alerts.

```
index=* ("suricata" OR "snort" OR "ET MALWARE" OR "ET TROJAN" OR "ATTACK_RESPONSE" OR "alert") host=("IDS1" OR "snort*") | table _time host _raw | sort _time
```

2. INITIAL ACCESS

9

Phishing — Browser from Email

WIN MAIL

Browser spawned by Thunderbird/Outlook.

```
index=* EventCode=1 ("thunderbird" OR "outlook") ("msedge" OR "chrome" OR "firefox" OR "iexplore") | table _time host User Image CommandLine ParentImage
```

Downloads Folder — New Files

WIN

Files landing in user Downloads. Training: l9k1JEYlHZ.exe in f.moore\Downloads.

```
index=* EventCode=11 "Downloads" | table _time host User Image TargetFilename Hashes | sort _time
```

Zone.Identifier (MOTW)

WIN

Internet-downloaded files.

```
index=* EventCode=11 "Zone.Identifier" | table _time host User TargetFilename | sort _time
```

Office Macro Child Procs

WIN

Office → cmd/PS = macro (.docm).

```
index=* EventCode=1 ("WINWORD" OR "EXCEL" OR "POWERPNT" OR "OUTLOOK") ("cmd.exe" OR "powershell" OR "wscript" OR "cscript" OR "mshta" OR "certutil") | table _time host User Image CommandLine ParentImage
```

Malicious LNK / Shortcut

WIN

Training: 432.lnk → BumbleBee. Use CyberChef "From Hex" to decode LNK content.

```
index=* (".lnk" OR "shortcut") EventCode=1 | table _time host User Image CommandLine ParentImage Hashes
```

ALL Known Malicious Files

ALL

Every known bad filename from all sources.

```
index=* ("security_update.exe" OR "dbstatus.exe" OR "logwatcher.exe" OR "db_exfil.exe" OR "tickets.exe" OR "creds.exe" OR "ps_exec.exe" OR "powerview.ps1" OR "dwm.exe" OR "bits.exe" OR "linpeas" OR "SharPersist" OR "horilla.zip" OR "J_James_Tax_Return" OR ".docm" OR "l9k1JEYlHZ.exe" OR "db_update.exe" OR "image_slider.exe" OR "healthcheck.exe" OR "SharpHound.ps1" OR "software_updater.ps1" OR "6.exe" OR "432.lnk" OR "minerxmr.exe" OR "nc.exe" OR "stage.bat" OR "fodshell_443.exe" OR "prst_servshell443.exe" OR "HID.dll" OR "nightmare.dll" OR "lockbit.exe" OR "7.exe" OR "IOBit.exe" OR "ws.phtml" OR "load.ps1" OR "fodhelper_load.ps1" OR "Setup.dll" OR "Mischa.dll" OR "QUADAGENT" OR "stolen_id_rsa") | table _time host Image CommandLine TargetFilename Hashes | sort _time
```

Web Shell Access

WEB LINUX

Ch3: ws.phtml. Training: stage.bat in wp-admin.

```
index=* ("cmd=" OR "cmd%3D" OR ".phtml" OR ".php?" OR ".asp?" OR ".jsp?" OR "webshell" OR "c99" OR "r57" OR "b374k" OR "stage.bat") ("200" OR "POST") | table _time host _raw | sort _time
```

SSH Brute Force / Login

LINUX

Training: alice/bob/wendy brute forced via SSH on linux01.

```
index=* ("Accepted password" OR "Accepted publickey" OR "Failed password") ("ssh" OR "sshd") | table _time host _raw |
```

```
sort _time
```

WebDAV Remote Mount

WIN

Training: WebDAV via net use for BumbleBee (webdav.4shared.com, lasex69621@cohodl.com).

```
index=* ("webdav" OR "4shared" OR "DavWWWRoot" OR "cohodl.com" OR "net use" "http") | table _time host User Image Comm
andLine
```

3. EXECUTION

7

PS -ep bypass

WIN

Execution policy bypass.

```
index=* EventCode=1 ("-ep bypass" OR "-ExecutionPolicy Bypass" OR "-exec bypass") | table _time host User CommandLine
ParentImage
```

Encoded PowerShell

WIN

Base64 stagers/C2. Use Revoke-Obfuscation to deobfuscate.

```
index=* EventCode=1 ("-enc " OR "-EncodedCommand" OR "FromBase64" OR "ToBase64" OR "[Convert]::" OR "base64") | table
_time host User CommandLine ParentImage Hashes
```

Certutil Abuse

WIN DC

Download/decode.

```
index=* "certutil" ("-urlcache" OR "-decode" OR "-encode" OR "-decodehex" OR "-f ") | table _time host User CommandLin
e Image ParentImage
```

LOLBins (wmic/bitsadmin/mshta/rundll32/fodhelper)

WIN

Living off the land.

```
index=* EventCode=1 ("wmic" OR "bitsadmin" OR "mshta" OR "regsvr32" OR "rundll32" OR "cmstp" OR "msiexec" OR "fodhelpe
r") | table _time host User CommandLine Image ParentImage Hashes
```

EXE from Users\ via PS/CMD Parent

WIN

Binaries executed from user dirs via PowerShell or CMD — classic malware execution pattern.

```
index=* EventCode=1 ParentImage="*powershell*" OR ParentImage="*cmd.exe*" | where like(Image, "%Users%") | where NOT l
ike(Image, "%OneDrive%") | table _time host User ParentImage Image CommandLine Hashes | sort _time
```

Linux Reverse Shell

LINUX WEB

Reverse shell payloads. Training: nc.exe in C:\Windows\Temp\.

```
index=* ("/dev/tcp/" OR "bash -i" OR "nc -e" OR "ncat" OR "mkfifo" OR "python -c" OR "python3 -c" OR "import socket" O
R "import pty" OR "perl -e" OR "exec(r.read())" OR "base64 -d | bash") | table _time host _raw | sort _time
```

Process Create 4688 (Legacy)

WIN

If no Sysmon.

```
index=* EventCode=4688 | table _time host SubjectUserName NewProcessName CommandLine ParentProcessName | sort _time
```

4. PERSISTENCE

19

schtasks /create

WIN

Process creation.

```
index=* EventCode=1 "schtasks" "create" | table _time host User CommandLine ParentImage
```

Task API (4698)

WIN

Task via Windows API.

```
index=* EventCode=4698 | table _time host SubjectUserName TaskName TaskContent
```

Known Bad Task Names

WIN

WindowsUpdate, UpdateHealthCheck, sw_update, bits.

```
index=* ("WindowsUpdate" OR "UpdateHealthCheck" OR "sw_update" OR "schtaskbackdoor") (EventCode=4698 OR EventCode=1) |
table _time host TaskName CommandLine
```

Files in \Windows\Tasks\

WIN DB

Stealth drop. Training: healthcheck.exe, SharpHound.ps1 in Tasks.

```
index=* EventCode=11 ("\\Windows\\Tasks\\" OR "\\System32\\Tasks\\") | table _time host Image TargetFilename Hashes
```

New Account (4720)

WIN DC

helpdesk_1, support_388945a0, adm1n.

```
index=* EventCode=4720 | table _time host SubjectUserName TargetUserName
```

Registry Run Keys (EC 13)

WIN

Training: prst_runshell443.exe in Run keys.

```
index=* EventCode=13 ("CurrentVersion\\Run" OR "RunOnce") | table _time host Image TargetObject Details User
```

New Service (7045)

WIN DC

bits, PSEXESVC, prst_servshell443.

```
index=* EventCode=7045 | table _time host ServiceName ServiceFileName AccountName
```

Service Install (4697 / sc create)

WIN

Training: LocalSystem = PrivEsc via service.

```
index=* (EventCode=4697 OR ("sc" "create") OR ("sc" "config")) "LocalSystem" | table _time host ServiceName CommandLin
e Image
```

Startup Folder

WIN

Autorun drops.

```
index=* EventCode=11 "Startup" | table _time host Image TargetFilename Hashes
```

SharPersist

WIN DC

Ch3: renamed net.exe on DC01.

```
index=* ("SharPersist" OR "schtaskbackdoor" OR "-t schtask" OR "-t service" OR "-m add") | table _time host Image Comm
andLine ParentImage
```

Linux Persistence

LINUX

Cron, systemd, .bashrc/.profile backdoors. Training: bob's .bashrc/.profile backdoor.

```
index=* ("crontab" OR "/etc/cron" OR "systemctl" OR ".service" OR ".timer" OR "/etc/systemd" OR ".bashrc" OR ".profile
" OR "/var/spool/cron") | table _time host _raw | sort _time
```

WMI Event Subscription (Sysmon 19/20/21)

WIN

The "holy grail" of stealth persistence. __EventFilter + CommandLineEventConsumer + FilterToConsumerBinding = SYSTEM persistence across reboots.

```
index=* (EventCode=19 OR EventCode=20 OR EventCode=21) | table _time host User EventCode Operation EventType Name Cons
```

```
umer Filter Destination
```

IFEO Debugger Hijack (sethc/utilman/osk)

WIN

Image File Execution Options\<target>.exe\Debugger → SYSTEM shell from lock screen (Sticky Keys attack).

```
index=* EventCode=13 TargetObject="*Image File Execution Options*" TargetObject="*Debugger" | table _time host Image T
argetObject Details User
```

Winlogon Shell/Userinit Hijack

WIN

Userinit accepts comma-separated list — attacker appends. Shell replace runs at logon before explorer.

```
index=* EventCode=13 (TargetObject="*Winlogon\\Userinit" OR TargetObject="*Winlogon\\Shell" OR TargetObject="*Winlogon
\\Taskman" OR TargetObject="*Winlogon\\Notify" OR TargetObject="*Winlogon\\AppSetup") | table _time host Image TargetO
bject Details
```

LSA Provider Hijack (lsass loads attacker DLL)

WIN DC

Authentication Packages / Security Packages / Notification Packages under HKLM\...\Lsa — DLL loaded by lsass at boot.

```
index=* EventCode=13 (TargetObject="*Lsa\\Security Packages" OR TargetObject="*Lsa\\Authentication Packages" OR Target
Object="*Lsa\\Notification Packages" OR TargetObject="*OSConfig\\Security Packages") | table _time host Image TargetOb
ject Details
```

COM Hijack (HKCU\Software\Classes\CLSID)

WIN

HKCU shadows HKLM for COM class resolution. Frequently-loaded COM object redirected to attacker DLL. No admin needed.

```
index=* EventCode=13 TargetObject="HKU*\\Software\\Classes\\CLSID*\\InprocServer32*" | table _time host Image TargetOb
ject Details User
```

BITS Persistence with SetNotifyCmdLine

WIN

BITS jobs persist across reboots via qmgr*.dat. /setnotifycmdline = run command when transfer completes.

```
index=* EventCode=1 Image="*bitsadmin.exe" (CommandLine="*setnotifycmdline*" OR CommandLine="*SetMinRetryDelay*" OR Co
mmandLine="*/transfer*http*") | table _time host User CommandLine
```

Hidden Scheduled Task (SD Deletion)

WIN

Attacker deletes SD value under TaskCache — schtasks no longer enumerates but task still fires. HAFNIUM tech.

```
index=* EventCode=13 TargetObject="*TaskCache\\Tree*" TargetObject="*SD" EventType="DeleteValue" | table _time host Im
age TargetObject
```

GPO Modification (T1484.001)

DC

Malicious GPO with startup script or scheduled task pushes to every machine in OU on 90-min refresh.

```
index=* (EventCode=5136 OR EventCode=5137) (ObjectClass="groupPolicyContainer" OR ObjectDN="*CN=Policies,CN=System*")
| table _time host SubjectUserName ObjectDN AttributeLDAPDisplayName OperationType
```

5. PRIVILEGE ESCALATION

12 ▼

Special Privs (4672)

WIN DC

Admin logon.

```
index=* EventCode=4672 | table _time host SubjectUserName PrivilegeList | sort _time
```

UAC Bypass (fodhelper/ms-settings)

WIN

Training: fodhelper.exe + fodhelper_load.ps1. Elevates from Medium to High IL.

```
index=* ("fodhelper" OR "ms-settings\\Shell\\Open\\command" OR "fodhelper_load" OR "eventvwr" OR "sdclt" OR "computerd
efaults") | table _time host Image CommandLine TargetObject Details
```

Unquoted Service Path

WIN

Training: IOBit.exe in C:\Program Files (x86)\IObit\. IUService.exe unquoted.

```
index=* EventCode=1 ("IOBit.exe" OR "C:\\Program.exe" OR "IUService") | table _time host Image CommandLine ParentImage
User
```

DLL Hijacking

WIN

Training: HID.dll in microsoft shared\ink\ for osk.exe.

```
index=* EventCode=11 (".dll") ("\\microsoft shared\\" OR "\\System32\\" OR "\\SysWOW64\\") NOT ("\\WinSxS") | table _t
ime host Image TargetFilename Hashes
```

PrintNightmare

WIN

nightmare.dll in offsec\AppData\Local\Temp → spoolsv → creates adm1n.

```
index=* ("nightmare" OR "spoolsv" OR "AddPrinterDriverEx" OR "CVE-2021-1675" OR "CVE-2021-34527") | table _time host I
mage CommandLine TargetFilename
```

Linux PrivEsc Tools

LINUX

linpeas, pspy, sudo, SUID. Training: root on web01 via command injection.

```
index=* ("linpeas" OR "LinPEAS" OR "linenum" OR "linux-exploit-suggester" OR "pspy" OR "sudo -l" OR "SUID" OR "cap_set
uid" OR "getcap" OR "/etc/shadow" OR "pkexec" OR "polkit") | table _time host _raw | sort _time
```

User → Admin Group

WIN DC

Added to privileged groups.

```
index=* (EventCode=4728 OR EventCode=4732 OR EventCode=4756) | table _time host SubjectUserName MemberName TargetUserN
ame GroupName
```

BYOVD — Vulnerable Driver Load (Sysmon 6)

WIN

Signed vulnerable drivers loaded to kill EDR: dbutil, gdrv, kprocesshacker, RTCore64, mhyprot, Procexp152, etc.

```
index=* EventCode=6 (ImageLoaded="*dbutil*" OR ImageLoaded="*gdrv*" OR ImageLoaded="*kprocesshacker*" OR ImageLoaded="
*RTCore64*" OR ImageLoaded="*mhyprot*" OR ImageLoaded="*Procexp152*" OR ImageLoaded="*Zemana*" OR ImageLoaded="*GMER*"
OR ImageLoaded="*IOBitUnlock*" OR Signature="" OR SignatureStatus!="Valid") | table _time host ImageLoaded Signature S
ignatureStatus Hashes
```

Zerologon (CVE-2020-1472)

DC

Mass 4742 where TargetUserName = DC computer account ($) with SubjectUserName = ANONYMOUS / empty = Netlogon exploit.

```
index=* EventCode=4742 TargetUserName="*$" (SubjectUserName="ANONYMOUS*" OR SubjectUserName="-" OR SubjectUserName="")
| stats count by host TargetUserName SubjectUserName | sort - count
```

NoPac / sAMAccountName Spoof (CVE-2021-42278/42287)

DC

Computer account created, then renamed to match a DC's sAMAccountName without the $. Watch 4741 → 4742.

```
index=* (EventCode=4741 OR EventCode=4742) | stats values(EventCode) as events values(TargetUserName) as targets by Su
bjectUserName host _time | where match(targets, "DC") AND match(events, "4741")
```

Potato-Family SYSTEM Escalation

WIN

Service account (IIS APPPOOL, NETWORK SERVICE) spawning High/System IL shell = JuicyPotato/PrintSpoofer/GodPotato/EfsPotato.

```
index=* EventCode=1 (User="*IIS APPPOOL*" OR User="*NETWORK SERVICE*" OR User="*LOCAL SERVICE*") (Image="*cmd.exe" OR
Image="*powershell.exe") (IntegrityLevel="System" OR IntegrityLevel="High") | table _time host User Image ParentImage
IntegrityLevel CommandLine
```

SeImpersonate/SeDebug Assigned to Non-Admin

WIN

4672 with SeImpersonate or SeDebug on unusual accounts — Potato precondition / Mimikatz precondition.

```
index=* EventCode=4672 (PrivilegeList="*SeImpersonatePrivilege*" OR PrivilegeList="*SeDebugPrivilege*" OR PrivilegeLis
```

```
t="*SeTcbPrivilege*" OR PrivilegeList="*SeLoadDriverPrivilege*") NOT (SubjectUserName="SYSTEM" OR SubjectUserName="LOC
AL SERVICE" OR SubjectUserName="NETWORK SERVICE" OR SubjectUserName="*$") | table _time host SubjectUserName Privilege
List
```

6. DEFENSE EVASION

8

AMSI Bypass

WIN

Training: AmsiScanBuffer patch or context overwrite with null bytes in PS.

```
index=* EventCode=4104 ("AmsiUtils" OR "AmsiScanBuffer" OR "amsi.dll" OR "bypass" "amsi" OR "AmsiContext") | table _ti
me host Message
```

Process Masquerading

WIN

Legit names from wrong paths. Training: check OriginalFileName vs Image mismatch.

```
index=* EventCode=1 ("\\Temp\\" OR "\\Tasks\\" OR "\\Downloads\\" OR "\\AppData\\" OR "\\ProgramData\\") (".exe") NOT
("\\Windows\\System32" OR "\\Windows\\SysWOW64" OR "\\Program Files") | table _time host Image CommandLine Hashes Pare
ntImage User
```

File Deletion / Cleanup

WIN

Evidence destruction.

```
index=* EventCode=4104 ("del " OR "Remove-Item" OR "rm " OR "erase " OR "Clear-EventLog") | table _time host Message
```

Event Log Clearing

WIN

Log tampering.

```
index=* (EventCode=1102 OR EventCode=104 OR "wevtutil" OR "Clear-EventLog") | table _time host _raw | sort _time
```

AV/EDR Tampering

WIN

Disabling Defender.

```
index=* ("Set-MpPreference" OR "DisableRealtimeMonitoring" OR "sc stop" OR "taskkill" "defender" OR "net stop") | tabl
e _time host _raw | sort _time
```

Timestomping (EC 2)

WIN

File time changed.

```
index=* EventCode=2 | table _time host Image TargetFilename CreationUtcTime PreviousCreationUtcTime
```

Non-Microsoft Binaries (Company Field)

WIN

Find unsigned / non-MS processes. Filters out VMware and known safe parents.

```
index=* EventCode=1 Company="*" Company!="Microsoft Corporation" Company!="Microsoft Corp." Company!="VMware, Inc." NO
T ParentImage="*svchost.exe" | table _time host User Image CommandLine Company Hashes | sort - _time
```

Network Events from Unusual Paths

WIN

Network connections from EXEs outside Program Files / System32 — likely malware beaconing.

```
index=* EventCode=3 NOT (Image="*Program Files*" OR Image="*Windows\\System32*" OR Image="*Windows\\SysWOW64*" OR Imag
e="*OneDrive*") | stats count by Image DestinationIp DestinationPort host | sort - count
```

7. CREDENTIAL ACCESS

23

Mimikatz / Cred Dump

WIN DB

Ch1: creds.exe. Training: healthcheck.exe OriginalFileName=mimikatz.exe.

```
index=* ("sekurlsa" OR "lsass" OR "procdump" OR "lazagne" OR "privilege::debug" OR "minidump" OR "logonpasswords" OR "
```

```
kerberos::list" OR "dpapi::" OR "mimikatz" OR "healthcheck.exe" OR "OriginalFileName" "mimikatz") | table _time host User CommandLine Image Hashes
```

LSASS Access (EC 10)

WIN

Direct LSASS memory access. Training: powershell.exe → lsass.exe.

```
index=* EventCode=10 "lsass.exe" | table _time host SourceImage GrantedAccess User
```

AS-REP Roasting (4768)

DC

No pre-auth.

```
index=* EventCode=4768 "Pre_Authentication_Type" "0x0" | table _time host Account_Name Client_Address Ticket_Encryption_Type
```

Kerberoasting (4769 burst)

DC

Burst of TGS requests.

```
index=* EventCode=4769 | stats count by IpAddress ServiceName TargetUserName TicketEncryptionType | sort - count
```

Kerberoasting — RC4 Filter

DC

0x17/0x18 = RC4-HMAC = strong Kerberoasting indicator.

```
index=* EventCode=4769 ("0x17" OR "0x18") | table _time host ServiceName TargetUserName IpAddress TicketEncryptionType
```

DCSync

DC

Replication GUIDs from non-DC.

```
index=* EventCode=4662 ("1131f6aa" OR "1131f6ad" OR "89e95b76") | table _time host SubjectUserName ObjectName Properties
```

NTDS.dit Dump

DC

Training: ntdsutil.exe process creation on DC.

```
index=* ("ntdsutil" OR "IFM" OR "ntds.dit" OR "vssadmin" "shadow") | table _time host Image CommandLine User
```

Clear-text Creds in CmdLine

WIN

Ch1: Qwerty09!, Password123.

```
index=* EventCode=1 ("-p " OR "password" OR "SecureString" OR "PSCredential" OR "ConvertTo-SecureString" OR "-Credential") | table _time host User CommandLine Image
```

Pass the Hash (LogonType 9)

WIN DC

NewCredentials logon = PtH. Often followed by PSEXESVC service install.

```
index=* EventCode=4624 "LogonType" "9" | table _time host TargetUserName SourceNetworkAddress LogonProcessName
```

Golden Ticket Detection

DC

Training: 10yr ticket lifetime, RC4 instead of AES, ticket user ≠ session user. Requires krbtgt hash.

```
index=* EventCode=4768 ("krbtgt" OR "0x17") | table _time host Account_Name Client_Address Ticket_Encryption_Type Ticket_Options
```

Pass the Ticket / TGT Injection

DC

Training: klist shows injected tickets. mary on CLIENT03 with TGT injection.

```
index=* ("klist" OR "kerberos::ptt" OR "Rubeus" OR "asktgt" OR "ptt" OR "kirbi") | table _time host CommandLine Image User
```

Linux Cred Harvest

LINUX

passwd, shadow, SSH keys. Training: stolen_id_rsa from /home/alice/.

```
index=* ("/etc/shadow" OR "/etc/passwd" OR ".ssh/id_rsa" OR ".ssh/authorized_keys" OR "stolen_id_rsa" OR "unshadow" OR "hashcat" OR "john") | table _time host _raw | sort _time
```

Net-NTLMv2 Auth Trigger Files

WIN

File types that can coerce Net-NTLMv2 auth requests to attacker SMB servers (.scf, .url, .rtf, .doc).

```
index=* (".rtf" OR ".scf" OR ".url" OR ".wmx" OR ".accdb" OR ".pub" OR ".asx") | table _time host TargetFilename Image User | sort _time
```

LSASS Dump — Suspicious Access Masks

WIN

Mimikatz-class dumping. Masks 0x1010/0x1410/0x1438/0x143A/0x1FFFFF = read virtual memory. Exclude known AV.

```
index=* EventCode=10 TargetImage="*lsass.exe" (GrantedAccess="0x1010" OR GrantedAccess="0x1410" OR GrantedAccess="0x1438" OR GrantedAccess="0x143a" OR GrantedAccess="0x1F0FFF" OR GrantedAccess="0x1FFFFF" OR GrantedAccess="0x1F3FFF") NOT (SourceImage="*MsMpEng*" OR SourceImage="*MsSense*" OR SourceImage="*CSFalcon*" OR SourceImage="*SentinelAgent*" OR SourceImage="*elastic-agent*") | table _time host SourceImage SourceUser GrantedAccess CallTrace
```

comsvcs.dll MiniDump (Signed LSASS Dump)

WIN

rundll32 comsvcs.dll MiniDump — one of the most common LSASS dump techniques.

```
index=* EventCode=1 Image="*rundll32.exe" CommandLine="*comsvcs*" CommandLine="*MiniDump*" | table _time host User ParentImage CommandLine
```

procdump / Task Manager LSASS Dump

WIN

procdump -ma lsass.exe / taskmgr right-click create dump. Look for dbgcore.dll/dbghelp.dll loaded by taskmgr.

```
index=* ((EventCode=1 (CommandLine="*procdump*lsass*" OR CommandLine="*-ma lsass*")) OR (EventCode=7 Image="*taskmgr.exe" (ImageLoaded="*dbgcore.dll" OR ImageLoaded="*dbghelp.dll"))) | table _time host User Image CommandLine ImageLoaded
```

Registry Hive Dump (SAM/SYSTEM/SECURITY)

WIN DC

reg save HKLM\SAM / HKLM\SYSTEM / HKLM\SECURITY → offline hash extraction with secretsdump.

```
index=* EventCode=1 ("reg save" OR "reg.exe save") (("HKLM\\SAM" OR "HKLM\\SYSTEM" OR "HKLM\\SECURITY") OR (" sam " OR " system " OR " security ")) | table _time host User CommandLine ParentImage
```

DCSync — Refined (Non-DC Principals)

DC

Replication GUIDs: 1131f6aa=Get-Changes, 1131f6ad=Get-Changes-All, 89e95b76=Get-Changes-In-Filtered-Set. Filter out sync accts.

```
index=* EventCode=4662 (Properties="*1131f6aa-9c07-11d1-f79f-00c04fc2dcd2*" OR Properties="*1131f6ad-9c07-11d1-f79f-00c04fc2dcd2*" OR Properties="*89e95b76-444d-4c62-991a-0facbeda640c*") NOT (SubjectUserName="*$" OR SubjectUserName="MSOL_*" OR SubjectUserName="AAD_*" OR SubjectUserName="ADSync*") | table _time host SubjectUserName SubjectDomainName ObjectName AccessMask Properties
```

DPAPI Abuse / Master Key Access

WIN

Browser cred theft, SharpDPAPI, DonPAPI. Watches Protect master keys + browser Login Data reads.

```
index=* (("AppData\\Roaming\\Microsoft\\Protect" OR "AppData\\Local\\Microsoft\\Credentials") OR "dpapi::" OR "SharpDPAPI" OR "DonPAPI" OR "Login Data" OR ("Local State" "chrome")) | table _time host User Image CommandLine TargetFilename
```

Shadow Credentials (msDS-KeyCredentialLink)

DC

Whisker/pyWhisker persistence: write key to msDS-KeyCredentialLink → PKINIT as target. Survives password reset.

```
index=* (EventCode=5136 OR EventCode=4662) ("msDS-KeyCredentialLink" OR "KeyCredentialLink" OR "Whisker" OR "pyWhisker") | table _time host SubjectUserName AttributeLDAPDisplayName ObjectDN AttributeValue
```

AdminSDHolder Modification (Persistent DA Access)

DC

Writing ACE to AdminSDHolder — SDProp propagates to protected groups every 60 min. Top-tier AD persistence.

```
index=* EventCode=5136 ObjectDN="*AdminSDHolder*" | table _time host SubjectUserName ObjectDN AttributeLDAPDisplayName
```

```
OperationType
```

krbtgt Password Change (Golden Ticket Mitigation)

DC

4724 on krbtgt = admin rotating (good) OR attacker preparing (bad). Normal orgs rotate rarely; every occurrence is investigation-worthy.

```
index=* (EventCode=4724 OR EventCode=4723) TargetUserName="krbtgt*" | table _time host SubjectUserName TargetUserName
```

8. DISCOVERY

6

PowerView / AD Enum

WIN DC

PowerView cmdlets.

```
index=* EventCode=4104 ("Get-NetUser" OR "Get-DomainUser" OR "Get-NetGroup" OR "Get-DomainGroup" OR "Get-NetComputer"
OR "Find-LocalAdminAccess" OR "Invoke-UserHunter" OR "PowerView" OR "Invoke-ShareFinder") | table _time host Message
```

BloodHound / SharpHound

WIN DC

Training: SharpHound.ps1 in Tasks, Invoke-Bloodhound on CLIENT4, 20240304105845_BloodHound.zip.

```
index=* ("SharpHound" OR "BloodHound" OR "Invoke-Bloodhound" OR "azurehound" OR "20240304105845_BloodHound") | table _
time host Image CommandLine TargetFilename | sort _time
```

Built-in Recon (net/whoami/klist)

WIN

Manual enumeration.

```
index=* EventCode=1 ("net user" OR "net group" OR "net localgroup" OR "whoami" OR "ipconfig" OR "systeminfo" OR "hostn
ame" OR "nltest" OR "dsquery" OR "quser" OR "arp -a" OR "netstat" OR "klist") | table _time host User CommandLine Pare
ntImage
```

AD Object Enum (4662)

DC

High volume = AD enum.

```
index=* EventCode=4662 | stats count by SubjectUserName ObjectType | sort - count
```

Linux Enum Commands

LINUX

Cloud metadata, cron, user enum.

```
index=* ("metadata.google" OR "169.254.169.254" OR "crontab -l" OR "cat /etc/passwd" OR "id" OR "uname -a" OR "ifconfi
g" OR "ip addr" OR "find / -perm") | table _time host _raw | sort _time
```

Account Lockout (4740)

WIN DC

After brute force.

```
index=* EventCode=4740 | table _time host TargetUserName TargetDomainName
```

9. LATERAL MOVEMENT

21

WinRM Sessions

WIN

New-PSSession, Enter-PSSession, Invoke-Command.

```
index=* EventCode=4104 ("New-PSSession" OR "Enter-PSSession" OR "Invoke-Command" OR "ComputerName") | table _time host
Message
```

WinRM File Drops (wsmprovhost)

WIN

Files via WinRM.

```
index=* EventCode=11 "wsmprovhost" | table _time host TargetFilename Hashes User
```

Copy-Item ToSession

WIN

File transfer via WinRM.

```
index=* EventCode=4104 ("Copy-Item" "ToSession") | table _time host Message
```

PsExec Service (7045)

WIN

PSEXESVC on target. Training: PtH → PsExec chain.

```
index=* EventCode=7045 ("PSEXESVC" OR "psexec") | table _time host ServiceName ServiceFileName AccountName
```

PsExec Child Procs

WIN

Procs launched by PsExec.

```
index=* EventCode=1 "PSEXESVC" | table _time host User Image CommandLine Hashes
```

Admin Share (C$/ADMIN$)

WIN

Admin share access.

```
index=* EventCode=5140 ("C$" OR "ADMIN$") | table _time host SubjectUserName ShareName IpAddress
```

Network Logon (Type 3)

WIN

SMB/WinRM/PsExec.

```
index=* EventCode=4624 "LogonType" "3" | table _time host TargetUserName TargetLogonId SourceNetworkAddress | sort _ti
me
```

RDP Logon (Type 10)

WIN

Training: h.johnson RDP → CLIENT3, DB1, FILE1. Also check SERVER04.

```
index=* EventCode=4624 "LogonType" "10" | table _time host TargetUserName SourceNetworkAddress
```

SMB (Port 445)

WIN

Network connections on SMB.

```
index=* EventCode=3 DestinationPort=445 | stats count by SourceIp DestinationIp Image User | sort - count
```

Net Use

WIN

SMB share mapping.

```
index=* EventCode=1 ("net use" OR "net1 use") | table _time host User CommandLine ParentImage
```

AnyDesk / Remote Tools

WIN

Training: AnyDesk, techinline for persistence/access.

```
index=* ("AnyDesk" OR "TeamViewer" OR "ScreenConnect" OR "ConnectWise" OR "RustDesk" OR "techinline" OR "unattended.te
chinline") | table _time host Image CommandLine User
```

SMB File Copy to UNC

WIN

Ch3: horilla.zip via UNC path.

```
index=* ("\\\\" "copy" OR "\\\\" "cp" OR "\\\\" "xcopy" OR "\\\\" "robocopy" OR "\\\\192.168") | table _time host _raw
| sort _time
```

DCOM Lateral (mmc/dllhost spawning shell)

WIN

MMC20.Application, ShellWindows, ShellBrowserWindow remote COM abuse. Parent=mmc.exe or dllhost.exe.

```
index=* EventCode=1 (ParentImage="*mmc.exe" OR ParentImage="*dllhost.exe") (Image="*cmd.exe" OR Image="*powershell.exe
" OR Image="*pwsh.exe") | table _time host User ParentImage ParentCommandLine Image CommandLine
```

Impacket wmiexec Signature (ADMIN$ Redirect)

WIN

wmiexec.py output redirect pattern: cmd /Q /c ... 2>&1 1> \\127.0.0.1\ADMIN$__<timestamp>.

```
index=* EventCode=1 ParentImage="*wmiprvse.exe" (CommandLine="*\\\\127.0.0.1\\ADMIN$\\__*" OR CommandLine="*2>&1*\\\\*
ADMIN$*") | table _time host User CommandLine
```

Cobalt Strike / PsExec Named Pipes (Sysmon 17/18)

WIN

CS SMB beacon: msagent_*, status_*, postex_*, MSSE-*-server. PsExec: PSEXESVC. Impacket: svcctl.

```
index=* (EventCode=17 OR EventCode=18) (PipeName="\\msagent_*" OR PipeName="\\status_*" OR PipeName="\\postex_*" OR Pi
peName="\\MSSE-*-server" OR PipeName="\\mojo.*" OR PipeName="\\PSEXESVC*" OR PipeName="\\PAExec*" OR PipeName="\\svcct
l") | table _time host Image User PipeName EventCode
```

Impacket atexec — Task Create+Run+Delete Cluster

WIN

atexec.py creates task, runs it, deletes within seconds — tight 4698+4699 cluster per source user.

```
index=* (EventCode=4698 OR EventCode=4699) | bin _time span=30s | stats values(EventCode) as events dc(EventCode) as e
vt_types by _time SubjectUserName host | where evt_types>=2
```

Lateral Velocity — One Source, Many Targets

WIN DC

Single user/IP authenticating to many hosts in short window = lateral fan-out or spray.

```
index=* EventCode=4624 LogonType=3 | bin _time span=10m | stats dc(host) as targets values(host) as hosts count by _ti
me TargetUserName IpAddress | where targets>=5 | sort - targets
```

Silver Ticket — 4624 Kerberos without DC 4769

DC

Forged TGS skips DC — auth arrives on target without preceding TGS request on any DC. Tune against renewals.

```
index=* EventCode=4624 LogonType=3 AuthenticationPackageName=Kerberos | eval lookup_key=TargetUserName."|".host | join
type=outer lookup_key [search index=* EventCode=4769 earliest=-15m | eval lookup_key=TargetUserName."|".ServiceName |
rename _time as kerb_ts | fields lookup_key kerb_ts] | where isnull(kerb_ts) | table _time host TargetUserName IpAddre
ss
```

RemoteInteractive to Non-Admin Workstation

WIN

RDP to end-user desktops from internal hosts — unusual in most environments.

```
index=* EventCode=4624 LogonType=10 host IN ("CLIENT*","WK*","WKSTN*") | stats count values(IpAddress) as src by Targe
tUserName host | sort - count
```

Explicit Credentials Logon (4648)

WIN

runas, PtH, remote admin with alt creds. TargetServerName reveals pivot destination.

```
index=* EventCode=4648 | table _time host SubjectUserName TargetUserName TargetServerName ProcessName
```

10. COMMAND & CONTROL

7

PS Downloads (iwr/wget/WebClient)

WIN

PowerShell downloads.

```
index=* EventCode=4104 ("Invoke-WebRequest" OR "iwr " OR "wget" OR "DownloadFile" OR "Start-BitsTransfer" OR "Net.WebC
lient" OR "DownloadString" OR "Invoke-RestMethod") | table _time host Message
```

Certutil Downloads

WIN DC

Download cradle.

```
index=* "certutil" "-urlcache" | table _time host CommandLine Image User
```

C2 / Malware Keywords

ALL

All known C2 frameworks + implants.

```
index=* ("Havoc" OR "Sliver" OR "meterpreter" OR "Metasploit" OR "Cobalt" OR "Beacon" OR "Empire" OR "Murdoc.exe" OR "
WinBackup.exe" OR "Murloc.exe" OR "BumbleBee" OR "QUADAGENT" OR "logwatcher.exe" OR "security_update.exe" OR "dbstatus
.exe" OR "lockbit" OR "akira") | table _time host Image CommandLine Hashes
```

DGA Domains (.life — BumbleBee)

ALL

Sysmon 22 DNS. Training: 12+ .life DGA domains. Cuckoo sandbox may reveal them when DNS fails (11001).

```
index=* (EventCode=22 OR "dns") ".life" | table _time host QueryName Image
```

Known C2 Domains (All)

ALL

All known malicious domains.

```
index=* ("techinline" OR "unattended.techinline" OR "lockbit-decryptor" OR "4shared.com" OR "x4z9arb.cn" OR "cohodl.co
m" OR "cmid1s1zeiu.life" OR "itszko2ot5u.life" OR "3v1n35i5kwx.life" OR "newdnq1xnl9.life" OR "jkyj6awt1ao.life" OR "d
drjv6y42b8.life" OR "1pnhp5o5za1.life" OR "y13iqvlfjl5.life" OR "xp0btfgegbo.life" OR "gpv3uw5tmy4.life" OR "5d7rdf3la
yn.life" OR "2aed6bvquxs.life") | table _time host _raw | sort _time
```

C2 Tasking (test.log)

WIN DB

Ch1: test.log C2 instructions.

```
index=* ("test.log" OR "type C:\\Windows\\Tasks") | table _time host _raw | sort _time
```

Linux C2 / Staging

LINUX WEB

wget, curl, python stagers, C2 ports. Training: 222.184.79.11:5329/5319.

```
index=* ("wget " OR "curl " OR "python3 -c" OR "python -c" OR "/dev/tcp" OR "exec(r.read())" OR "urlopen" OR ":8080/"
OR ":8000/" OR ":4444/" OR ":9001/" OR ":5329/" OR ":5319/") | table _time host _raw | sort _time
```

11. EXFILTRATION

6 ▼

Compression / Staging

WIN

Training: 20240304105845_BloodHound.zip in \Tasks\.

```
index=* ("Compress-Archive" OR ".zip" OR ".7z" OR ".rar" OR "tar " OR "makecab" OR "BloodHound.zip" OR "horilla.zip")
| table _time host _raw | sort _time
```

Exfil Binary → DB → Outbound

DB

Ch1: db_exfil.exe.

```
index=* EventCode=3 ("127.0.0.1" OR "exfil" OR "db_exfil") NOT ("svchost" OR "System" OR "sqlservr") | table _time hos
t Image DestinationIp DestinationPort User
```

SCP / SFTP Transfer

WIN

Training: scp.exe for exfil, user y8ki5kl5.

```
index=* ("scp.exe" OR "scp " OR "WinSCP" OR "sftp " OR "pscp" OR "FileZilla" OR "y8ki5kl5") | table _time host Image C
ommandLine DestinationIp Hashes
```

Outbound Transfer Patterns

ALL

Repeated outbound from same binary.

```
index=* EventCode=3 NOT (DestinationIp="10.*" OR DestinationIp="127.*" OR DestinationIp="192.168.*" OR DestinationIp="
0.0.0.0") | stats count values(DestinationPort) as ports by Image DestinationIp host | sort - count
```

SMB Exfil to Attacker

WIN

Ch3: SMB share copy.

```
index=* EventCode=3 DestinationPort=445 NOT (DestinationIp="10.*" OR DestinationIp="127.*") | table _time host Image S
ourceIp DestinationIp User
```

Cryptominer

WIN

Training: minerxmr.exe in C:\Windows\.

```
index=* ("minerxmr" OR "xmrig" OR "stratum" OR "monero" OR "cryptonight" OR "coinhive") | table _time host Image Comma
ndLine Hashes
```

12. RANSOMWARE DETECTION

7

Ransom Notes

ALL

akira_readme in diverse Verzeichnisse, lockbit, RECOVER.

```
index=* ("akira_readme" OR "lockbit" OR "RECOVER" OR "DECRYPT" OR "README" OR "YOUR_FILES" OR "HOW_TO") (".txt" OR ".h
tml" OR ".hta") | table _time host TargetFilename Image _raw | sort _time
```

Encrypted Extensions

ALL

.nmh (Akira), .lockbit, etc.

```
index=* (".nmh" OR ".lockbit" OR ".encrypted" OR ".crypt" OR ".enc" OR ".locked" OR ".petya" OR ".mischa" OR ".goldene
ye") | table _time host TargetFilename Image | sort _time
```

Akira Binaries

ALL

l9k1JEYlHZ.exe in f.moore\Downloads & C:\share\ALL\, renamed as db_update.exe / image_slider.exe.

```
index=* ("l9k1JEYlHZ" OR "db_update.exe" OR "image_slider.exe" OR "637E28B38086FF9EFD1606805FF57AAF6CDEC4537378F019D60
70A5EFDC9C983" OR "3c92bfc71004") | table _time host Image CommandLine TargetFilename Hashes
```

LockBit / Petya / Mischa / GoldenEye

ALL

Training: Setup.dll=Petya loader, Mischa.dll, core.dll/elevate*.dll=GoldenEye.

```
index=* ("lockbit.exe" OR "lockbit-decryptor" OR "Setup.dll" OR "Mischa.dll" OR "core.dll" OR "elevate_x86.dll" OR "el
evate_x64.dll" OR "goldeneye" OR "petya" OR "B240B6861889") | table _time host Image CommandLine TargetFilename Hashes
```

Mass File Mod (EC 11/23)

ALL

High volume creates/deletes = encryption.

```
index=* (EventCode=11 OR EventCode=23) | stats count by Image host | where count > 100 | sort - count
```

VSS Shadow Delete

ALL

Prevent recovery.

```
index=* ("vssadmin" "delete" OR "wmic" "shadowcopy" OR "bcdedit" "recoveryenabled" OR "wbadmin" "delete") | table _tim
e host CommandLine Image User
```

Ransomware via SMB Share

ALL

Training: Akira binary pushed to C:\share\ALL\ via SMB for lateral ransomware deployment.

```
index=* EventCode=11 "\\share\\" ("l9k1JEYlHZ" OR "akira" OR "lockbit" OR ".exe") | table _time host Image TargetFilen
ame Hashes
```

13. IOC HASH HUNTING

5

Challenge Hashes

ALL

Murdoc, WinBackup, Murloc, PowerView, Exfil, logwatcher.

```
index=* ("62ccdfad" OR "9be7df15" OR "3ccbc477" OR "50d6e979" OR "7ee5bd37" OR "bccb122c" OR "BDD3AD53") | table _time
host EventCode Image TargetFilename CommandLine Hashes
```

Training Hashes (All)

ALL

Akira, LockBit, Mimikatz, AnyDesk, BumbleBee, software_updater, 6.exe, 432.lnk, prst_servshell.

```
index=* ("637E28B3" OR "3c92bfc7" OR "B240B686" OR "61C0810A" OR "55E4CE3F" OR "5F78E94C" OR "557F6E27" OR "FE3C93DB"
OR "94A8EF7D" OR "3FB5CF71") | table _time host EventCode Image TargetFilename CommandLine Hashes
```

All Known Filenames

ALL

Every malicious filename.

```
index=* ("security_update.exe" OR "dbstatus.exe" OR "logwatcher.exe" OR "db_exfil.exe" OR "tickets.exe" OR "creds.exe"
OR "ps_exec.exe" OR "powerview.ps1" OR "dwm.exe" OR "bits.exe" OR "linpeas" OR "SharPersist" OR "healthcheck.exe" OR "
SharpHound.ps1" OR "software_updater.ps1" OR "l9k1JEYlHZ.exe" OR "minerxmr.exe" OR "nc.exe" OR "stage.bat" OR "nightma
re.dll" OR "HID.dll" OR "IOBit.exe" OR "lockbit.exe" OR "ws.phtml" OR "horilla.zip" OR "432.lnk" OR "Setup.dll" OR "Mi
scha.dll" OR "QUADAGENT" OR "stolen_id_rsa" OR "fodhelper_load.ps1" OR "load.ps1" OR "prst_servshell443.exe" OR "prst_
runshell443.exe") | table _time host EventCode Image TargetFilename CommandLine Hashes | sort _time
```

Hash Template (replace HASH_PREFIX)

ALL

First 8 chars of SHA256.

```
index=* "HASH_PREFIX" | table _time host EventCode Image TargetFilename CommandLine Hashes
```

NTLM Hash (Training)

ALL

Training: offsec NTLM 2892d26cdf84d7a70e2eb3b9f05c425e.

```
index=* ("2892d26cdf84d7a70e2eb3b9f05c425e") | table _time host _raw
```

⏱ 14. TIMELINE RECONSTRUCTION

6 ▼

Full Host Timeline

ALL

Replace TARGET_HOST. Correlate EC1+EC3 to find C2 beacons.

```
index=* host="TARGET_HOST" (EventCode=1 OR EventCode=3 OR EventCode=11 OR EventCode=4104) | eval act=case(EventCode==1
,"PROC: ".Image, EventCode==3,"NET: ".DestinationIp.":".DestinationPort, EventCode==11,"FILE: ".TargetFilename, EventC
ode==4104,"PS: ".substr(Message,1,200)) | table _time EventCode act Hashes | sort _time
```

Correlate by LogonId

WIN

Single session tracking.

```
index=* host="TARGET_HOST" (EventCode=1 OR EventCode=11) "0xLOGON_ID" | table _time EventCode Image TargetFilename Com
mandLine Hashes | sort _time
```

Suspicious Procs (Filtered)

WIN

Windows noise removed.

```
index=* host="TARGET_HOST" EventCode=1 NOT ("svchost.exe" OR "lsass.exe" OR "services.exe" OR "csrss.exe" OR "wininit.
exe" OR "smss.exe" OR "ctfmon.exe" OR "taskhostw.exe" OR "RuntimeBroker" OR "SearchUI" OR "sihost.exe" OR "conhost.exe
" OR "dllhost.exe") | table _time Image CommandLine Hashes User ParentImage | sort _time
```

Combined Logon Timeline

WIN

4776+4624+4634 with hostname/source.

```
index=* (EventCode=4776 OR EventCode=4624 OR EventCode=4634) | table _time host EventCode TargetUserName WorkstationNa
me Workstation SourceNetworkAddress | sort _time
```

First-Time User Profile

WIN

Detect rogue accounts first logon.

```
index=* EventCode=11 "svchost.exe" "Users" ("Desktop" OR "Documents" OR "Downloads") | stats earliest(_time) as first_
seen values(TargetFilename) as folders by host User
```

Time Range Filter Template

ALL

Narrow any query to a specific time window. Replace dates. Append to any query.

```
index=* earliest="07/24/2024:00:00:00" latest="07/25/2024:23:59:59" | table _time host _raw | sort _time
```

15. LINUX / WEB SERVER

5

Linux Sysmon Rex Parse

LINUX WEB

Extract XML fields. Replace WEB01.

```
index=* host="WEB01" "Sysmon" | rex field=_raw "Name=\"CommandLine\">(?<CmdLine>[^<]+)" | rex field=_raw "Name=\"Image
\">(?<Img>[^<]+)" | rex field=_raw "Name=\"User\">(?<Usr>[^<]+)" | rex field=_raw "Name=\"ParentImage\">(?<ParentImg>[
^<]+)" | rex field=_raw "Name=\"Hashes\">(?<Hash>[^<]+)" | table _time Usr Img CmdLine ParentImg Hash | sort _time
```

Linux LOTLbins

LINUX

All common Linux attacker tools.

```
index=* ("sudo" OR "chmod" OR "chown" OR "SUID" OR "find /" OR "linpeas" OR "pspy" OR "getcap" OR "passwd" OR "/tmp/"
OR "wget" OR "curl" OR "nc " OR "python" OR "reverse" OR "su " OR "cron" OR "pkexec") | table _time host _raw | sort _
time
```

Linux User Modification

LINUX

User/group creation.

```
index=* ("useradd" OR "adduser" OR "usermod" OR "groupadd" OR "chpasswd" OR "visudo") | table _time host _raw | sort _
time
```

Egress Busting

LINUX

Training: outbound probing on many ports to test firewall egress filtering. If blocked = good DiD.

```
index=* ("dev/tcp" OR "nc -zv" OR "nslookup" OR "dig ") | table _time host _raw | sort _time
```

DB — SQL/xp_cmdshell

DB

Training: dbhost 172.16.50.12.

```
index=* ("xp_cmdshell" OR "sp_oacreate" OR "EXEC master" OR "sqlcmd" OR "osql" OR "bcp " OR "openrowset") | table _tim
e host _raw | sort _time
```

16. WEB APPLICATION ATTACKS

5

SQL Injection

WEB DB

Training: UNION SELECT, PG_SLEEP, 1=1, ModSec id "942100".

```
index=* ("UNION SELECT" OR "UNION ALL" OR "OR 1=1" OR "PG_SLEEP" OR "sleep(" OR "benchmark(" OR "942100" OR "sqlmap" O
R "1=1--" OR "information_schema") | table _time host _raw | sort _time
```

Shellshock (CGI Injection)

WEB LINUX

Training: () { :;}; in headers, GET /cgi-bin/ with 200 response = successful exploit.

```
index=* ("() {" OR "() { :;}" OR "/cgi-bin/" OR "shellshock") | table _time host _raw | sort _time
```

LFI (Path Traversal)

WEB

Training: ../ traversal, wp-config.php inclusion.

```
index=* ("../" OR "..../" OR "..%2f" OR "wp-config.php" OR "/etc/passwd" OR "file=" OR "page=" OR "include=") ("200" O
R "GET" OR "POST") | table _time host _raw | sort _time
```

ModSecurity / WAF Alerts

WEB

ModSec rule IDs. Training: 942100=SQLi, 941100=XSS, 932100=RCE.

```
index=* ("ModSecurity" OR "modsec" OR "942100" OR "941100" OR "932100" OR "SecRule") | table _time host _raw | sort _time
```

Web Server Lateral (4624/4625 from webhost)

WEB

Training: Compromised web server used for lateral brute force. Check 172.16.51.33, 172.16.50.11.

```
index=* (EventCode=4624 OR EventCode=4625) ("172.16.51.33" OR "172.16.50.11") | table _time host TargetUserName IpAddress LogonType | sort _time
```

17. KERBEROS ATTACKS (Focused)

10

Kerberoast — RC4 Burst Per User

DC

5+ SPNs requested RC4 (0x17) from one src in 5 min = Kerberoast. Exclude $ accts and krbtgt.

```
index=* EventCode=4769 TicketEncryptionType="0x17" ServiceName!="*$" ServiceName!="krbtgt" | bin _time span=5m | stats dc(ServiceName) as spn_count values(ServiceName) as spns by _time TargetUserName IpAddress | where spn_count>=5 | sort - spn_count
```

AS-REP Roast (Pre-Auth Not Required)

DC

4768 with PreAuthType=0 → account has DONT_REQ_PREAUTH → hash crackable offline (hashcat 18200).

```
index=* EventCode=4768 PreAuthType="0" | stats count values(TargetUserName) as targets by IpAddress host | where count>=2 | sort - count
```

Golden Ticket — Abnormal Lifetime

DC

Forged TGTs often have 10-year lifetimes. Domain default is typically 10 hours.

```
index=* (EventCode=4768 OR EventCode=4769) | rex field=_raw "TicketLifetime[=:]\s*(?<lifetime>\d+)" | where lifetime > 36000 | table _time host TargetUserName ServiceName Client_Address lifetime
```

Kerberos RC4 in AES-Only Environment

DC

Modern AD should use AES (0x11/0x12). RC4 (0x17) = downgrade, Kerberoast, or overpass-the-hash.

```
index=* (EventCode=4768 OR EventCode=4769) (TicketEncryptionType="0x17" OR TicketEncryptionType="0x18") NOT ServiceName="krbtgt" | stats count by host TargetUserName ServiceName IpAddress TicketEncryptionType | sort - count
```

TGT Request (4768) Without User Interaction

DC

Rubeus asktgt / overpass-the-hash: TGT requested from non-interactive session.

```
index=* EventCode=4768 Result_Code="0x0" | stats count values(Client_Address) as src_ips by TargetUserName Ticket_Encryption_Type | where count>=3 AND Ticket_Encryption_Type="0x17" | sort - count
```

PKINIT Auth (Cert-Based — ESC1/Shadow Creds)

DC

Pre-auth type 15/16/17 = certificate authentication. Flag any unusual cert auth, especially to privileged accounts.

```
index=* EventCode=4768 (PreAuthType="15" OR PreAuthType="16" OR PreAuthType="17") | table _time host TargetUserName Client_Address CertIssuerName CertSerialNumber CertThumbprint
```

S4U2Self / S4U2Proxy Delegation Abuse

DC

Transited Services field populated = delegation used. Unexpected entries = RBCD / constrained delegation abuse.

```
index=* EventCode=4769 Transited_Services!="-" Transited_Services!="" | table _time host TargetUserName ServiceName Transited_Services Client_Address
```

Overpass-the-Hash Pattern

DC

TGT request with RC4 from host where user never normally logs on. Combine with Mimikatz sekurlsa::pth signatures.

```
index=* EventCode=4768 TicketEncryptionType="0x17" | stats earliest(_time) as first_seen count values(Client_Address)
as src_ips by TargetUserName | where first_seen > relative_time(now(),"-24h")
```

4771 Pre-Auth Failures — Brute / Spray

DC

0x18 = bad password. Cluster per source / per target indicates brute force or spray.

```
index=* EventCode=4771 Failure_Code="0x18" | bin _time span=10m | stats dc(TargetUserName) as users count as attempts
values(Client_Address) as src by _time | where users>=5 OR attempts>=20 | sort - attempts
```

Kerberos Ticket Tools (Rubeus/Mimikatz/ticket.kirbi)

DC

Rubeus asktgt/ptt/kerberoast + Mimikatz kerberos:: module + .kirbi tickets.

```
index=* (EventCode=1 OR EventCode=4104) ("Rubeus" OR "kerberos::ptt" OR "kerberos::golden" OR "kerberos::list" OR "ask
tgt" OR "asktgs" OR "kerberoast " OR ".kirbi" OR "tgtdeleg" OR "s4u") | table _time host User CommandLine Image Messag
e
```

18. ADCS / CERTIFICATE ABUSE

9 ▼

Certipy / Certify Tool Detection

DC

Main ADCS abuse tools. Certipy (Python), Certify (.NET). Watch command lines and PS 4104.

```
index=* ("certipy" OR "Certify.exe" OR "Certify find" OR "Certify request" OR "Certify pfx" OR "PassTheCert" OR "Forge
Cert" OR "Rubeus asktgt*/cert*") | table _time host User Image CommandLine Message
```

Certificate Issued — SAN Contains UPN

DC

CertSvc 4886/4887: issued cert. If SubjectAltName contains UPN of a privileged user = ESC1 abuse.

```
index=* source="*CertSvc*" (EventCode=4886 OR EventCode=4887 OR EventCode=4898) (Attributes="*upn=*" OR SubjectAltName
="*@*" OR Subject="*DA*" OR Subject="*Admin*") | table _time host Requester Subject SubjectAltName TemplateName Attrib
utes
```

Certificate Template Modification (ESC4)

DC

Write to template object = ESC4. Precursor to ESC1 setup or direct abuse.

```
index=* EventCode=5136 ObjectDN="*CN=Certificate Templates,CN=Public Key Services*" | table _time host SubjectUserName
ObjectDN AttributeLDAPDisplayName OperationType AttributeValue
```

Certificate Enrolled by Low-Priv User

DC

Low-priv account enrolling certs on sensitive templates = ESC1/ESC2/ESC3 indicator.

```
index=* source="*CertSvc*" EventCode=4886 NOT (Requester="*admin*" OR Requester="*$" OR Requester="SYSTEM") | stats co
unt values(TemplateName) as templates by Requester host | where count>=1
```

Certificate Request with manageCA / ESC7

DC

ESC7: ManageCA/ManageCertificates rights abuse to issue / approve pending.

```
index=* source="*CertSvc*" (EventCode=4882 OR EventCode=4885) | table _time host User Message
```

CA Flag Change — EDITF_ATTRIBUTESUBJECTALTNAME2 (ESC6)

DC

EDITF flag enables SAN override on any request. Modification of CA policy = near-game-over.

```
index=* ("EDITF_ATTRIBUTESUBJECTALTNAME2" OR "certutil*-setreg*policy*EditFlags" OR "CA policy") | table _time host Us
er CommandLine Message
```

PKINIT Auth from New/Unusual Host

DC

Cert-based logon (4768 PreAuth=16) from host that doesn't normally auth this user = stolen cert / shadow creds.

```
index=* EventCode=4768 PreAuthType="16" | stats earliest(_time) as first_seen values(Client_Address) as clients by Tar
getUserName | where first_seen > relative_time(now(),"-24h")
```

ADCS Web Enrollment (ESC8 Target)

DC

HTTP POST to /certsrv/certfnsh.asp — the NTLM relay target for ESC8. Unauthenticated HTTP path on CA = vulnerable.

```
index=* ("/certsrv/" OR "certfnsh.asp" OR "certrqad.asp" OR "/certsrv/certfnsh.asp") | table _time host src clientip u
rl http_method http_status
```

Cert Template Enumeration (LDAP)

DC

LDAP queries to CN=Certificate Templates,CN=Public Key Services,CN=Services,CN=Configuration = Certipy find / recon.

```
index=* EventCode=1644 (SearchFilter="*pKIEnrollmentService*" OR SearchFilter="*pKICertificateTemplate*" OR ObjectDN="
*CN=Certificate Templates*") | table _time host ClientIP User SearchFilter
```

19. NTLM RELAY / COERCION

8

PetitPotam / EFSRPC Coercion (MS-EFSR)

DC

Coerce DC auth via EFSRPC (EfsRpcOpenFileRaw, EfsRpcEncryptFileSrv). Named pipe: \\pipe\lsarpc or \\pipe\efsrpc.

```
index=* EventCode=5145 (RelativeTargetName="lsarpc" OR RelativeTargetName="efsrpc" OR RelativeTargetName="efsr") | tab
le _time host ShareName RelativeTargetName SubjectUserName IpAddress
```

PrinterBug / SpoolSample (MS-RPRN)

DC

RpcRemoteFindFirstPrinterChangeNotification coercion. Target \\pipe\spoolss.

```
index=* EventCode=5145 RelativeTargetName="spoolss" | table _time host ShareName SubjectUserName IpAddress
```

DFSCoerce / ShadowCoerce (MS-DFSNM / MS-FSRVP)

DC

Coercion via DFS namespace mgmt or File Server VSS pipes.

```
index=* EventCode=5145 (RelativeTargetName="netdfs" OR RelativeTargetName="FssagentRpc") | table _time host RelativeTa
rgetName SubjectUserName IpAddress
```

LLMNR / NBT-NS Response from Unusual Host

WIN

Responder-style poisoning: host claims to be every requested name on 5355/udp (LLMNR) or 137/udp (NBT-NS).

```
index=* EventCode=3 (DestinationPort=5355 OR DestinationPort=137 OR SourcePort=5355 OR SourcePort=137) | stats count d
c(DestinationIp) as tgt by SourceIp Image | where tgt>=5 | sort - count
```

NTLMv1 Usage (Should Never Happen)

DC

NTLMv1 is trivially crackable. 4624 with PackageName indicating v1.

```
index=* EventCode=4624 (PackageName="NTLM V1" OR LmPackageName="NTLM V1") | table _time host TargetUserName IpAddress
WorkstationName LogonProcessName PackageName
```

NTLM Relay Pattern — Mismatched Source

DC

4624 NTLM where WorkstationName (Windows-reported) differs from computer-account source = relayed auth.

```
index=* EventCode=4624 AuthenticationPackageName="NTLM" LogonType=3 | eval ws_upper=upper(WorkstationName) | eval src_
host=upper(replace(IpAddress,"\.\d+$","")) | where WorkstationName!="" AND NOT match(TargetUserName, ws_upper) | table
_time host TargetUserName WorkstationName IpAddress
```

Outbound NTLM from DC (Coercion Success Indicator)

DC

DC computer account (DC$) authenticating outbound = likely coerced victim.

```
index=* EventCode=4648 SubjectUserName="*$" TargetServerName!="" | stats count values(TargetServerName) as targets by
host SubjectUserName | where count>=1
```

Coercion Tools (Coercer, PetitPotam, DFSCoerce)

DC

Public coercion scripts. Watch CLI and PS 4104.

```
index=* ("PetitPotam" OR "Coercer" OR "DFSCoerce" OR "ShadowCoerce" OR "PrinterBug" OR "SpoolSample" OR "EfsRpcOpenFil
eRaw" OR "RpcRemoteFindFirstPrinterChangeNotification") | table _time host User CommandLine Image Message
```

●⛓ 20. KILL-CHAIN CORRELATION (Multi-Tactic)

8 ▼

Host Traversing ≥4 Tactics in 24h

ALL

Tags every qualifying event with a tactic, then groups per host. Host hitting many tactics = active intrusion.

```
index=* earliest=-24h | eval tactic=case(
  (EventCode=1 AND (match(ParentImage,"(?i)winword|excel|powerpnt|outlook|onenote"))),"initial_access",
  (EventCode=4104 AND match(_raw,"(?i)(IEX|DownloadString|FromBase64|amsiscanbuffer|invoke-mimikatz)")),"execution",
  (EventCode=7045 OR EventCode=4698 OR (EventCode=13 AND match(TargetObject,"(?i)run|winlogon|image file execution")))
,"persistence",
  (EventCode=10 AND TargetImage="*lsass.exe"),"credential_access",
  (EventCode=4662 AND match(Properties,"1131f6aa|1131f6ad")),"cred_access_dcsync",
  (EventCode=4688 AND match(CommandLine,"(?i)(net group|net view|nltest|whoami|systeminfo)")),"discovery",
  (EventCode=4624 AND LogonType="3" AND AuthenticationPackageName="NTLM"),"lateral_ntlm",
  (EventCode=4624 AND LogonType="10"),"lateral_rdp",
  (match(_raw,"(?i)vssadmin.*delete|wbadmin.*delete") OR EventCode=1102),"defense_evasion",
  (match(_raw,"(?i)rclone|7z.*-p|mega(sync|cmd)")),"exfiltration",
  true(),null()) | where isnotnull(tactic) | stats dc(tactic) as tactics_hit values(tactic) as chain by host | where t
actics_hit>=4 | sort - tactics_hit
```

LSASS Access → Lateral Auth Within 10 min

WIN

Cred theft immediately followed by lateral using harvested creds. Classic smash-and-grab signature.

```
index=* EventCode=10 TargetImage="*lsass.exe" (GrantedAccess="0x1010" OR GrantedAccess="0x1410" OR GrantedAccess="0x14
38" OR GrantedAccess="0x143a" OR GrantedAccess="0x1F0FFF" OR GrantedAccess="0x1FFFFF") | eval lsass_time=_time | eval
source_host=host | fields source_host lsass_time SourceUser | join type=inner source_host [search index=* EventCode=46
24 (LogonType=3 OR LogonType=10) | eval source_host=IpAddress | eval lat_time=_time | eval tgt_host=host | fields sour
ce_host lat_time TargetUserName tgt_host] | where lat_time > lsass_time AND (lat_time - lsass_time) < 600 | table lsas
s_time lat_time source_host tgt_host TargetUserName
```

Ransomware-Imminent (Exfil + VSS Delete + Svc Stop)

ALL

Correlated signals that precede mass encryption by minutes. ≥2 triggers = PAGE SOC immediately.

```
index=* earliest=-6h | eval signal=case(
  match(_raw,"(?i)(rclone|mega(sync|cmd)|7z.*-p.*-mhe)"),"exfil_tool",
  match(_raw,"(?i)vssadmin.*delete.*shadow"),"vss_delete",
  match(_raw,"(?i)wbadmin.*delete.*catalog"),"backup_delete",
  match(_raw,"(?i)bcdedit.*recoveryenabled.*no"),"boot_tamper",
  (EventCode=7036 AND match(_raw,"(?i)(sqlserver|exchange|veeam|backup|mysql|oracle)")),"db_svc_stop",
  match(_raw,"(?i)net\s+stop.*(sql|exchange|veeam|backup)"),"svc_kill_cmd",
  (EventCode=1102),"log_clear",
  true(),null()) | where isnotnull(signal) | stats dc(signal) as signals values(signal) as triggers values(CommandLine
) as cmds by host | where signals>=2 | sort - signals
```

PetitPotam Coerce → DCSync (Relay Attack)

DC

RPC coerce pipe access + DCSync from same host/user within minutes = B4/B6 attack chain executing.

```
(index=* EventCode=5145 RelativeTargetName IN ("lsarpc","efsrpc","netdfs","spoolss")) OR (index=* EventCode=4662 (Prop
erties="*1131f6aa-9c07-11d1-f79f-00c04fc2dcd2*" OR Properties="*1131f6ad-9c07-11d1-f79f-00c04fc2dcd2*")) | eval tag=if
(EventCode=5145,"coerce_rpc","dcsync") | bin _time span=10m | stats values(tag) as tags earliest(_time) as first lates
t(_time) as last values(SubjectUserName) as users values(IpAddress) as ips by host _time | where mvcount(tags)>=2
```

BloodHound Collection + Kerberoast Same User

DC

LDAP/SMB session enum burst + RC4 4769 flood from same principal within 1h = B3 chain.

```
index=* ((EventCode=4769 TicketEncryptionType="0x17") OR (EventCode=5145 RelativeTargetName IN ("netlogon","lsarpc","s
amr","netsessenum")) OR (EventCode=4104 AND match(_raw,"(?i)(SharpHound|Invoke-BloodHound|Invoke-Kerberoast|GetUserSPN
s)"))) | bin _time span=1h | stats dc(EventCode) as evt_types values(EventCode) as events count by _time TargetUserNam
e IpAddress | where evt_types>=2 AND count>=10
```

Phish → Payload → C2 Within 1 Hour

WIN

Office macro parent → suspicious child → external network connection, all on the same host within 60 min.

```
index=* EventCode=1 (ParentImage="*winword.exe" OR ParentImage="*excel.exe" OR ParentImage="*powerpnt.exe" OR ParentIm
age="*outlook.exe" OR ParentImage="*onenote.exe") | eval phish_time=_time | fields host User phish_time ParentImage |
join host [search index=* EventCode=3 NOT (DestinationIp="10.*" OR DestinationIp="172.16.*" OR DestinationIp="172.17.*
" OR DestinationIp="192.168.*" OR DestinationIp="127.*") (Image="*powershell.exe" OR Image="*rundll32.exe" OR Image="*
regsvr32.exe" OR Image="*mshta.exe" OR Image="*cmd.exe") | eval c2_time=_time | fields host c2_time DestinationIp Dest
inationPort Image] | where c2_time > phish_time AND (c2_time - phish_time) < 3600 | table phish_time c2_time host User
ParentImage Image DestinationIp
```

New Service + DA Group Change Same Host/Hour

WIN DC

Service install + protected group modification in same 1h window = lateral deployment + persistence.

```
index=* earliest=-1h (EventCode=7045 OR EventCode=4697 OR EventCode=4728 OR EventCode=4732 OR EventCode=4756) | eval k
ind=case(EventCode IN (7045,4697),"svc_install",EventCode IN (4728,4732,4756),"group_add",true(),null()) | stats dc(ki
nd) as k values(kind) as kinds values(ServiceName) as svcs values(MemberName) as members by host | where k>=2
```

Discovery Velocity + Next-Step Auth

ALL

Host runs >=8 distinct discovery commands and then makes outbound auth — lateral prep in progress.

```
index=* EventCode=4688 (NewProcessName="*net.exe" OR NewProcessName="*nltest.exe" OR NewProcessName="*whoami.exe" OR N
ewProcessName="*systeminfo.exe" OR NewProcessName="*tasklist.exe" OR NewProcessName="*quser.exe" OR NewProcessName="*w
mic.exe" OR NewProcessName="*dsquery.exe" OR NewProcessName="*setspn.exe" OR NewProcessName="*nbtstat.exe") | bin _tim
e span=10m | stats dc(NewProcessName) as cmd_variety count by _time host SubjectUserName | where cmd_variety>=8 | join
host [search index=* EventCode=4624 LogonType=3 | stats count by host] | table _time host SubjectUserName cmd_variety
count
```

QUICK REFERENCE

Exam Tips (Training + Cheatsheet Consolidated)

- Sysmon EC1 (ProcCreate) + EC3 (NetConnect) correlation → find C2 beacons (Empire, BumbleBee)
- OriginalFileName mismatch → masqueraded binary (e.g. healthcheck.exe OriginalFileName=mimikatz.exe)
- CyberChef "From Hex" → decode .lnk shortcut content to reveal embedded downloader commands
- Cuckoo sandbox DNS failure (11001) → isolate DGA domain strings from sandbox output
- Don't isolate hosts during active IR unless ransomware is encrypting — monitor first
- Ensure PS Module Logging (4103), Script Block Logging (4104), Transcription Logging via GPO
- Golden Ticket: 10yr lifetime, RC4 (not AES), no preceding 4768 for user — klist mismatch
- Silver Ticket: 4624 Kerberos on target WITHOUT preceding 4769 on any DC
- Kerberoast signature: 4769 EType 0x17 for ≥5 SPNs from one user in short window
- DCSync: 4662 Properties 1131f6aa (Get-Changes) or 1131f6ad (Get-Changes-All) from non-DC
- ADCS ESC1: cert enrollment with SAN=DA-UPN → 4768 with PreAuthType 16 (PKINIT)
- PetitPotam/PrinterBug: EID 5145 with RelativeTargetName lsarpc/efsrpc/spoolss
- BYOVD: Sysmon EID 6 for known vulnerable drivers (dbutil/gdrv/RTCore64/mhyprot)
- LSASS dump: Sysmon EID 10 to lsass.exe with GrantedAccess 0x1010/0x1410/0x1438/0x1F0FFF
- NTLM Relay: 4624 NTLM where WorkstationName doesn't match the TargetUser's normal host
- Egress busting on many ports → if blocked, good Defense-in-Depth (DMZ/VLAN/egress filtering)
- Always check Logon Type first on 4624 — Type 3 = lateral, Type 9 = PtH, Type 10 = RDP
- FILE1 = departmental SMB shares, TICKETS1 = Jira, WIKI1 = Confluence, DEV1 = defensive team

FILE WRITER → METHOD

EC 11 Image	Method
svchost.exe	SMB
wsmprovhost.exe	WinRM
PSEXESVC.exe	PsExec
powershell.exe	PS download
explorer.exe	RDP/GUI

KEY EVENT CODES

EC	What
1	Process Create
2	File Time Changed

3	Network Connect
10	Process Access (lsass)
11	File Create
13	Registry Set
22	DNS Query
23	File Delete
4104	PS Script Block
4624	Logon (3/9/10)
4625	Failed Logon
4662	AD Object Access
4688	Process (legacy)
4697	Service Install
4698	Task Created
4720	Account Created
4740	Account Lockout
4768	AS-REQ (ASREP/GoldenTkt)
4769	TGS-REQ (0x17=Kerbroast)
4776	NTLM (hostname!)
5140	Share Access
7045	Service Installed
1102	Log Cleared

ATTACKER IPS (ALL SOURCES)

IP	Src	Role
192.168.50.235	Ch1	Staging
192.168.50.11	Ch1	Exfil C2
192.168.12.231	Ch2	Attacker
192.168.12.60	Ch3	Attacker
192.168.50.63	Tr	C2 Akira/Mimikatz
192.168.50.67	Tr	BumbleBee DL
192.168.51.50	Tr	Kali (attacker01)
192.168.51.54	Tr	attacker02
192.229.211.108	Tr	LockBit C2
222.184.79.11	Tr	Jenkins/WebLogic C2
226.211.90.236	Tr	Cyware feed
172.16.51.33	Tr	Compr. webserver

COMPROMISED USERS (ALL)

User	Src	How
n.harris	Ch1	Phishing WK3
m.smith	Ch1	Cracked cred
helpdesk_1	Ch1	Rogue account
m.thomas	Ch2	Compromised
m.williams	Ch2	Compromised
h.johnson	Tr	DomAdmin RDP lateral
e.taylor	Tr	BumbleBee LNK
t.allen	Tr	PS downloader
f.moore	Tr	Ransomware
offsec	Tr	PtH/PrivEsc

support_388945a0	Tr	APT3 persistence
adm1n	Tr	PrintNightmare
Administrator	Tr	Brute force/hashdump
alice/bob/wendy	Tr	SSH brute force
mary	Tr	TGT injection
root	Tr	Cmd injection web01

KNOWN HOSTS (ALL)

Host	Role	Src
WK01-WK3	Workstation	Ch
DB1	Database (DMZ)	Ch/Tr
DC01/DC1	Domain Ctrl	Ch/Tr
WEB01/web01	Web Server	Ch/Tr
APPSRV01/appsrv01	App Server	Ch/Tr
FILE1	File/SMB	Tr
MAIL1	Mail Server	Tr
CLIENT1-6	Workstations	Tr
IDS1/snort01-03	IDS/Router	Tr
HRIS	HR System	Ch3
CUCKOO	Sandbox	Tr
TICKETS1	Jira	Tr
WIKI1	Confluence	Tr
DEV/DEV1	Defense Team	Tr
SPLUNK/SPLUNK1	SIEM	Tr
SERVER01/04	Win Server	Tr
linux01/02	Web/Linux	Tr
CLIENT01/03	Endpoints	Tr
dbhost(172.16.50.12)	Database	Tr
webhost(172.16.50.11)	Web/FTP	Tr

PRIVESC TECHNIQUES

Technique	Indicator
UAC Bypass	fodhelper + ms-settings reg
Unquoted Path	IOBit.exe / C:\Program.exe
DLL Hijack	HID.dll in ink\ for osk.exe
PrintNightmare	spoolsv → nightmare.dll → adm1n
Golden Ticket	10yr ticket, RC4, user mismatch
DCSync	4662 + repl GUIDs
NTDS.dit	ntdsutil / vssadmin shadow
Service LocalSystem	4697 + LocalSystem account

RANSOMWARE FAMILIES

Family	Ext	Note / Loader
Akira	.nmh	akira_readme.txt
LockBit	.lockbit	lockbit-decryptor.com
Petya	MBR	Setup.dll (loader)
Mischa	various	Mischa.dll (payload)
GoldenEye	various	core.dll + elevate*.dll

KEY MALICIOUS FILES

File	Function
security_update.exe	C2 Implant
healthcheck.exe	Mimikatz
SharpHound.ps1	AD Enum (Tasks)
l9k1JEYlHZ.exe	Akira Ransom
lockbit.exe	LockBit
minerxmr.exe	Cryptominer
nightmare.dll	PrintNightmare
HID.dll	DLL Hijack
IOBit.exe	Unquoted Path
432.lnk	BumbleBee Drop
nc.exe	Reverse Shell
stage.bat	WordPress Loader
ws.phtml	Web Shell
Setup.dll	Petya Loader
Mischa.dll	Mischa Payload
QUADAGENT	PS C2 Utility
stolen_id_rsa	SSH Key (alice)
prst_servshell443	Service Persist
prst_runshell443	Run Key Persist
fodhelper_load.ps1	UAC Bypass

DGA DOMAINS (.LIFE)

Domain
cmid1s1zeiu.life
itszko2ot5u.life
3v1n35i5kwx.life
newdnq1xnl9.life
jkyj6awt1ao.life
ddrjv6y42b8.life
1pnhp5o5za1.life
y13iqvlfjl5.life
xp0btfgegbo.life
gpv3uw5tmy4.life
5d7rdf3layn.life
2aed6bvquxs.life

LOGON TYPES (4624/4625)

Type	Meaning / Signal
2	Interactive (console)
3	**Network** — SMB/WMI/WinRM/PsExec
4	Batch — scheduled task
5	Service
7	Unlock
8	**NetworkCleartext** — basic auth
9	**NewCredentials** — runas /netonly, PtH
10	RemoteInteractive — RDP

11	CachedInteractive

KERBEROS ETYPE / PRE-AUTH

Code	Meaning
0x11	AES128 (modern)
0x12	AES256 (modern)
0x17	**RC4-HMAC — Kerberoast/OPtH signal**
0x18	RC4-EXP (downgrade)
Pre-Auth Type (4768)	
0	**None — AS-REP roast**
2	PA-ENC-TIMESTAMP (normal)
15/16/17	**PKINIT — cert auth (ESC1/ShadowCred)**
4771 Failure Codes	
0x6	User unknown
0x12	Account disabled/locked
0x18	**Bad password — brute force**

LSASS ACCESS MASKS (SYSMON 10)

Mask	Meaning
0x1000	Query limited — benign
0x1010	**Read memory — Mimikatz**
0x1410	**Read + query — Mimikatz**
0x1438	**Dump read**
0x143A	**Full read**
0x1F0FFF	**All access — dumping**
0x1FFFFF	**All access — dumping**
0x40	DUP_HANDLE — handle theft

ADCS ESC QUICK MAP

ID	Vulnerability
ESC1	Template: SAN supplied + client auth EKU
ESC2	Any-Purpose EKU / no EKU
ESC3	Enrollment Agent template abuse
ESC4	Weak ACLs on template
ESC6	CA flag EDITF_ATTRIBUTESUBJECTALTNAME2
ESC7	ManageCA / ManageCertificates rights
ESC8	NTLM relay to CA web enrollment
ESC9/10	Cert mapping weakness
ESC11	RPC encryption not enforced
ESC13	OID group misconfig

PARENT → CHILD RED FLAGS

Parent	Bad Child
winword/excel/outlook	cmd/ps/rundll32/mshta
w3wp.exe	cmd/ps/net/whoami
nginx/apache2	sh/bash/curl/wget
wmiprvse.exe	cmd/ps (WMI remote)
wsmprovhost.exe	any (WinRM session)
services.exe	unknown binary

`mmc.exe / dllhost.exe`	cmd/ps (DCOM lateral)
`lsass.exe`	**never spawn children**
`spoolsv.exe`	cmd/ps = PrintNightmare

SYSMON EIDS QUICK

EID	Use
`1`	Process create (+cmdline)
`3`	Network connection
`6`	Driver load — BYOVD
`7`	Image load — DLL sideload
`8`	CreateRemoteThread — injection
`10`	ProcessAccess — LSASS dump
`11`	File create — webshell, staging
`12/13`	Registry create / set
`15`	Zone.Identifier (MOTW)
`16`	Sysmon config change — tamper
`17/18`	Named pipe — CS/PsExec
`19/20/21`	WMI filter/consumer/binding
`22`	DNS query
`25`	Process hollowing

CRITICAL EVENT IDS (TOP PRIORITY)

EID	Signal
`1102`	**Security log cleared**
`104`	System log cleared
`4719`	**Audit policy changed**
`4662`	DCSync (with repl GUIDs)
`4672`	Special privs assigned
`4697/7045`	Service installed
`4698/4702`	Task create/update
`4720`	New account
`4728/4732/4756`	Member added to sec group
`4742`	Computer acct changed (Zerologon)
`5136`	AD object modified
`5145`	Detailed share access

EXFIL DESTINATIONS / TOOLS

Signature	What
`*.mega.nz, gfs.mega`	MEGA cloud exfil
`rclone.conf`	Rclone config
`--transfers --bwlimit`	Rclone args
`7z -p -mhe=on`	Encrypted archive
`anonfiles, gofile.io`	Anonymous upload
`wetransfer, bashupload`	File sharing
`transfer.sh, 0x0.st`	Ephemeral upload
`discord webhook`	Discord C2/exfil
`api.telegram.org`	Telegram bot C2
`New-MailboxExportRequest`	Exchange mailbox exfil

TOOL → FINGERPRINT

Tool	Artifact
PsExec	PSEXESVC svc + pipe
Impacket psexec.py	Random .exe on ADMIN$
Impacket wmiexec.py	\\127.0.0.1\ADMIN$__ts
Impacket atexec.py	Task create+run+del <30s
Cobalt Strike SMB	msagent_/status_/postex_/MSSE- pipes
Mimikatz	LSASS EID 10 GA 0x1010/1410
Rubeus	4769 RC4 burst, ptt, asktgt
SharpHound/BloodHound	LDAP+SMB session burst
Certipy/Certify	Cert enrollment anomaly
Responder	LLMNR/NBT-NS replies
ntlmrelayx	NTLM src/WS mismatch
PetitPotam	lsarpc/efsrpc pipe hits

CHAIN ORDER (MEMORIZE)

#	Chain
B1	Phish→RCE→PS→C2→LSASS→Lateral→Ransom
B2	ExposedWeb→Webshell→Potato→LSASS→DA
B3	LowUser→SPNEnum→Kerberoast→Crack→Lateral
B4	Coerce→NTLMRelay→LDAP/ADCS→DCSync
B5	ADCS ESC1→Cert with SAN=DA→PKINIT
B6	PetitPotam→Relay to CA web→Cert→DC
B10	Zerologon→DC$ pwd=empty→DCSync
B11	DCSync→krbtgt hash→Golden
B13	Silver: no DC 4769 for the TGS
B14	ShadowCred: msDS-KeyCredentialLink
B16	BYOVD driver→kernel RW→EDR kill

🚨 TOP-PRIORITY HUNTS (5-MIN TRIAGE)

#	Query / Signal
1	EID 1102 — Security log cleared
2	Sysmon 10 → lsass with 0x1xxx masks
3	EID 4662 Properties 1131f6aa/ad
4	4769 RC4 burst ≥5 SPNs in 5 min
5	w3wp/nginx spawning shell
6	Office spawning ps/cmd/rundll32
7	vssadmin delete shadows
8	New svc EID 7045 w/ random name
9	New task EID 4698 w/ ps/cmd action
10	Logon Type 9 (runas /netonly)
11	Driver load Sysmon 6 (BYOVD)
12	WMI subscription EID 19/20/21